

Docling PDF Content Support Matrix

A consolidated technical reference detailing element extraction, layout reconstruction, and multimodal pipeline support in Docling.

PDFs can contain diverse elements ranging from layout structures and raster graphics to interactive forms, annotations, and low-level object streams. The consolidated reference below outlines how **Docling** handles each content type across deep learning, OCR, and vision-language modules.

CONTENT CATEGORY	SPECIFIC PDF ELEMENT	DOCLING SUPPORT	IMPLEMENTATION / MECHANISM IN DOCLING
CORE TEXT & TYPOGRAPHY			
Text Hierarchy	Body Text & Paragraphs	Full	Extracted into structured <code>TextItem</code> nodes with preserved reading order.
Text Hierarchy	Headings & Hierarchy	Full	Detects <code>TITLE</code> , <code>SECTION_HEADER</code> , etc., mapping them to standard Markdown <code>#</code> , <code>##</code> levels.
Lists & Items	Lists & Bullet Points	Full	Identifies <code>LIST_ITEM</code> markers (numbered/bulleted) and preserves list indentation.
Layout Flow	Multi-Column Text	Full	Uses vision-based layout analysis (DocLayNet) to resolve human reading flow across column splits.
TABULAR DATA			
Tables	Simple & Complex Grid Tables	Full	TableFormer neural network reconstructs spanning cells, headers, and column/row grids.
Tables	Borderless / White-Space Tables	Full	Reconstructed via structural layout cues even when ruling lines are absent.
VISUALS & GRAPHICS			
Bitmap Images	Raster Images & Photographs	Full	Detects bounding boxes as <code>PictureItem</code> ; crops and exports via <code>generate_picture_images</code> (PIL format).
Vector Art	Vector Graphics & Line Art	Full	Rasterized and treated as figures/drawings with bounding-box extraction.
Visual Analytics	Charts & Infographics	Full / VLM	Classified as pictures; can generate textual interpretations using integrated Vision-Language Models (<code>do_picture_description</code>).
TECHNICAL & SCIENTIFIC			
Mathematics	Math Formulas (Inline & Display)	Full	Converted directly into LaTeX format (<code>\$. . . \$</code> and <code>\$\$. . . \$\$</code>) via <code>do_formula_enrichment</code> .

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Source Code	Code Blocks & Scripts	Full	Isolates monospaced snippets (<code>CodeItem</code>), preserves whitespace, and detects programming language.
FORMS & CONTROLS			
Form Fields	Key-Value Pairs	Full	Detects label-to-value spatial bindings (<code>FIELD_KEY</code> , <code>FIELD_VALUE</code>).
Form Inputs	Checkboxes & Radio Buttons	Full	Classifies state as <code>CHECKBOX_SELECTED</code> or <code>CHECKBOX_UNSELECTED</code> .
AcroForms	AcroForms / Interactive Forms	Partial	Parses visible rendered field values; underlying raw AcroForm interactive dictionaries are flattened.
NAVIGATION & REFERENCE			
Page Artifacts	Document "Furniture"	Full	Tags <code>PAGE_HEADER</code> and <code>PAGE_FOOTER</code> as non-body artifacts, filtering noise from RAG pipelines.
Citations	Footnotes & References	Full	Detects citations and associates footnote anchors with target text at the page base.
Hyperlinks	Hyperlinks & Cross-References	Full	Preserves URI targets and internal document anchors within Markdown/HTML export.
Structure	Document Outlines / Bookmarks	Full	Ingests PDF Table of Contents (TOC) trees to enhance hierarchical document sectioning.
SCANNED & OCR DATA			
Scans	Scanned / Degraded Pages	Full	Passes bitmaps through OCR backends (EasyOCR, Tesseract, RapidOCR, Mac OCR, or KServe V2).
Machine Codes	Barcodes & QR Codes	Partial	Bounded as pictures/visual elements; decoding encoded string payloads requires an external reader (e.g., <code>pyzbar</code>).
DOCUMENT LAYERS & METADATA			
Properties	Document Metadata	Full	Extracts standard metadata (author, title, creation date, producer, page count) into document properties.
Watermarks	Watermarks & Stamps	Ignored / Filtered	Layout analysis separates background watermark artifacts from primary semantic text flow.
Annotations	Annotations & Sticky Notes	Partial	Visual annotations rendered on the canvas are read via layout/OCR; internal PDF comment annotation trees are typically bypassed.
ADVANCED & BINARY OBJECTS			

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OCG Layers	Layers (OCGs)	Partial	Flattens visible PDF optional content groups into the rendered page view.
Attachments	Embedded File Attachments	No	Embedded binary file streams in the PDF object catalog are not automatically unpacked.
Rich Media	Multimedia & 3D Objects	No	Embedded audio, video clips, and 3D meshes (U3D/PRC) are not extracted.
Security	Encrypted / Password Protected	Conditional	Requires decrypting the file with the password prior to conversion.

SUPPORT LEVEL CLASSIFICATION KEY

Full	Native end-to-end extraction and semantic parsing	Full / VLM	Vision-Language Model multi-modal enrichment
Partial	Rendered visual extraction / flattened support	Conditional	Supported subject to prerequisite authentication
Ignored	Filtered out by design to prevent noise	No	Not parsed from object streams